

N100 Financial Intelligence Platform Analyst Guide

Streamlit Analytics • Financial Screener • Peer Comparison • PDF Tear Sheets •
FastAPI

Day 44 Documentation

Version 1.0

This guide explains how an analyst can use the N100 platform to screen companies, inspect financial profiles, compare peers, review sectors, generate tear sheets, and consume the platform through its REST API.

1. Platform Overview

The N100 Financial Intelligence Platform is an analyst-oriented financial research application built around a normalized SQLite database, an analytics layer, a Streamlit user interface, and a FastAPI service layer.

The platform combines company fundamentals, financial ratios, stock prices, market-cap information, peer groups, sector information, analysis text, pros and cons, and derived analytics into a single workflow.

Core analyst capabilities

- Search and open a company profile using a ticker or company name.
- Review revenue, profit, profitability, leverage, valuation, and cash-flow indicators.
- Run a multi-factor financial screener using adjustable thresholds.
- Use predefined screener presets to quickly explore research filters.
- Compare a company with its peer group using normalized metrics and radar-chart data.
- Review sector-level statistics and company membership.
- Generate and download pre-generated PDF company tear sheets.
- Access company, screener, sector, peer, valuation, and health information through FastAPI.

Architecture: Data sources → ETL / cleaning → SQLite → analytics → FastAPI + Streamlit → analyst outputs.

The database is **data/nifty100.db**; the API is mounted under **/api/v1**; generated tear sheets are stored under **reports/tearsheets/**.

2. Getting Started

Project root: C:\Users\reka\OneDrive\Desktop\N100

Running Streamlit:

```
streamlit run app.py
```

Use the actual Streamlit entry file present in the repository if it differs from this example.

Running FastAPI:

```
uvicorn src.api.main:app --reload
```

The FastAPI application is exposed through **src/api/main.py**.

Regression validation:

```
pytest -q
```

The Day 44 final regression suite completed with **171 passed in 27.07 seconds**.

Formatting: Black was run across **src/** and **tests/**; 60 files were reformatted and 14 were unchanged.

3. Streamlit Dashboard Navigation

Company Profile

- Search by ticker or company name.
- Review company identity, sector, and key financial indicators.
- Inspect revenue and net-profit trends.
- Review ROE and ROCE trends.
- Read available pros and cons.

Financial Screener

- Adjust financial thresholds.
- Use a preset to populate filters.
- Review the result count and ranked results.
- Download filtered results as CSV.

Peer Comparison

- Select a company or peer group.
- Review comparable companies and percentile information.
- Use radar visualization to compare company, peer average, and benchmark.

Sector / analytics views

Use sector screens to understand company counts and median metrics before interpreting an individual company.

4. Using the Financial Screener

The screener is intended for systematic first-pass research. It supports ten financial dimensions through configurable filters and includes six presets.

Typical workflow

- Open the Screener screen.
- Begin with a preset when appropriate.
- Adjust individual thresholds.
- Review result count.
- Inspect the returned metrics.
- Open company profiles or peer comparisons for follow-up.
- Download filtered results as CSV.

Supported API filters

Minimum ROE, maximum debt-to-equity, minimum free cash flow, sector, minimum five-year revenue CAGR, minimum five-year PAT CAGR, and maximum P/E.

Interpretation

A screener result is a filtered research universe, not a conclusion about a company. Interpret results with sector context, data coverage, valuation, business quality, and qualitative information.

Treat null values as unavailable data rather than zero.

5. Company Profile Analysis

The Company Profile page consolidates information an analyst normally needs before moving into peer or valuation analysis.

Company search

Use the autocomplete/search control to locate a company. ABB and TCS were used during project validation.

KPI review

- Review company identifiers and sector classification.
- Check profitability and leverage indicators.
- Review valuation information where available.
- Review free-cash-flow and growth measures where populated.

Historical charts

Revenue and Net Profit charts provide a time-series view of business scale and earnings. ROE and ROCE charts provide a profitability-efficiency view.

Pros and Cons

The qualitative section summarizes available source analysis. Important claims should be validated against underlying financial data.

Data-quality note

Some derived fields can be null when source coverage or normalization does not provide a reliable value.

6. Peer Comparison and Sector Analysis

Peer analysis places a company's metrics in context. The platform includes peer groups and percentile calculations across multiple financial dimensions.

Peer workflow

- Select the company.
- Identify the relevant peer group.
- Review peer membership and percentile ranks.
- Compare the company with peer averages and benchmark values.
- Use the radar visualization to identify areas for deeper investigation.

Peer percentile metrics

The platform computes percentile-based comparisons for ten metrics. Percentiles represent relative positions within the selected comparison universe.

Sector analysis

The sector API provides company counts and median ROE, P/E, and debt-to-equity. The validated API returned 11 sectors.

Analyst practice

Use sector and peer context before interpreting a single metric because leverage, margins, capital intensity, and valuation ranges can differ across industries.

7. PDF Tear Sheets

The platform generates company tear sheets as PDF files for offline review and sharing. Pre-generated tear sheets are stored under **reports/tearsheets/**.

How to use a tear sheet

- Select a company with an available tear sheet.
- Download or open the generated PDF.
- Use it for compact review of fundamentals and analytics.
- Cross-check important values against the live profile when necessary.

Coverage considerations

Not every company necessarily has identical historical coverage. Companies with insufficient financial history can be skipped during tear-sheet generation.

Recommended review

Before relying on a tear sheet, verify ticker, reporting period, data coverage, and any null or unavailable metrics.

8. FastAPI Analyst Guide

The REST API provides programmatic access to the analytical universe. It is mounted under **/api/v1**.

Health

GET /api/v1/health

The validated health response reports status, database row counts, uptime, and API version. The project previously returned status **ok** and version **1.0.0**.

Company endpoints

- GET /api/v1/companies
- GET /api/v1/companies/{ticker}
- GET /api/v1/companies/{ticker}/pl
- GET /api/v1/companies/{ticker}/bs
- GET /api/v1/companies/{ticker}/cashflow
- GET /api/v1/companies/{ticker}/ratios
- GET /api/v1/companies/{ticker}/tearsheet

Screening and comparison

- GET /api/v1/screener
- GET /api/v1/sectors
- GET /api/v1/sectors/{sector}/companies
- GET /api/v1/peers/{group_name}
- GET /api/v1/companies/{ticker}/peers/compare
- GET /api/v1/market-cap/{ticker}

9. API Examples and Analyst Workflows

Company search

```
curl "http://127.0.0.1:8000/api/v1/companies?search=ABB"
```

Screener

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_de=1&max_pe=30"
```

Sector

```
curl "http://127.0.0.1:8000/api/v1/sectors"
```

Company profile workflow

- Identify the ticker using the company list.
- Request the company profile.
- Request ratios and historical statements.
- Request peer comparison data.
- Request the tear sheet when PDF output is required.

Recommended research sequence

Universe → Screener → Company Profile → Sector Context → Peer Comparison → Valuation → Tear Sheet.

10. Data, Database, and Output Reference

Primary database: data/nifty100.db

Key tables:

companies — company master and identifiers

analysis — source analysis / qualitative text

balancesheet — balance-sheet history

cashflow — cash-flow history

profitandloss — P&L; history

financial_ratios — derived financial ratios

market_cap — market-cap / valuation information

peer_groups — peer-group membership

peer_percentiles — relative metric rankings

sectors — sector classifications

stock_prices — historical price data

prosandcons — qualitative pros and cons

documents — document metadata / source records

Important output locations:

reports/tearsheets/ — generated company PDF tear sheets

output/ — parser, performance, review, and other generated outputs

docs/ — project documentation

data/raw/ — raw Excel source files

data/supporting/ — supporting financial and market datasets

11. Quality Checks, Troubleshooting, and Best Practices

Regression testing

Run `pytest -q` from the project root. The Day 44 baseline is **171 passed in 27.07s**.

Import-path troubleshooting

If `pytest` reports **ModuleNotFoundError: No module named 'src'**, verify that the current directory is the N100 project root. A temporary Windows CMD fix is:

```
set PYTHONPATH=%CD%  
pytest -q
```

Best practices

- Confirm reporting period before comparing historical metrics.
- Treat null values as unavailable data, not zero.
- Use peer and sector context before interpreting ratios.
- Check data coverage for companies with short histories.
- Use the API health endpoint when diagnosing service availability.
- Run the regression suite after code or dependency changes.
- Keep generated outputs separate from raw source data.

Day 44 end state: documentation, formatting, and regression validation completed successfully.

12. Quick Analyst Checklist

Before research

- Confirm Streamlit and API availability.
- Check API health when using the API.
- Confirm sector, peer group, and reporting period.

During screening

- Select appropriate preset or thresholds.
- Review result count.
- Inspect multiple metrics.
- Export CSV when required.

During company review

- Review latest KPIs.
- Inspect revenue and profit history.
- Inspect ROE / ROCE history.
- Review qualitative pros and cons.
- Check peer and sector context.
- Review valuation information.

Before sharing output

- Confirm ticker and company name.
- Confirm data period and coverage.
- Check missing values.
- Generate or open the relevant tear sheet.
- Record the API endpoint or exported CSV when reproducibility matters.

Appendix — Day 44 Validation Record

The Day 44 documentation milestone was followed by project-wide formatting and regression validation.

Formatting result

Black completed successfully: **60 files reformatted** and **14 files left unchanged**.

Regression result

The complete pytest suite completed successfully with **171 passed in 27.07 seconds**, with zero collection errors, failures, or test errors in the final run.

Release readiness checklist

- Documentation PDF created and stored in the docs directory.
- Source and test files formatted with Black.
- Full automated test suite passing.
- API and Streamlit workflows documented.
- Tear-sheet and analyst workflows documented.

N100 Financial Intelligence Platform — Analyst Guide v1.0